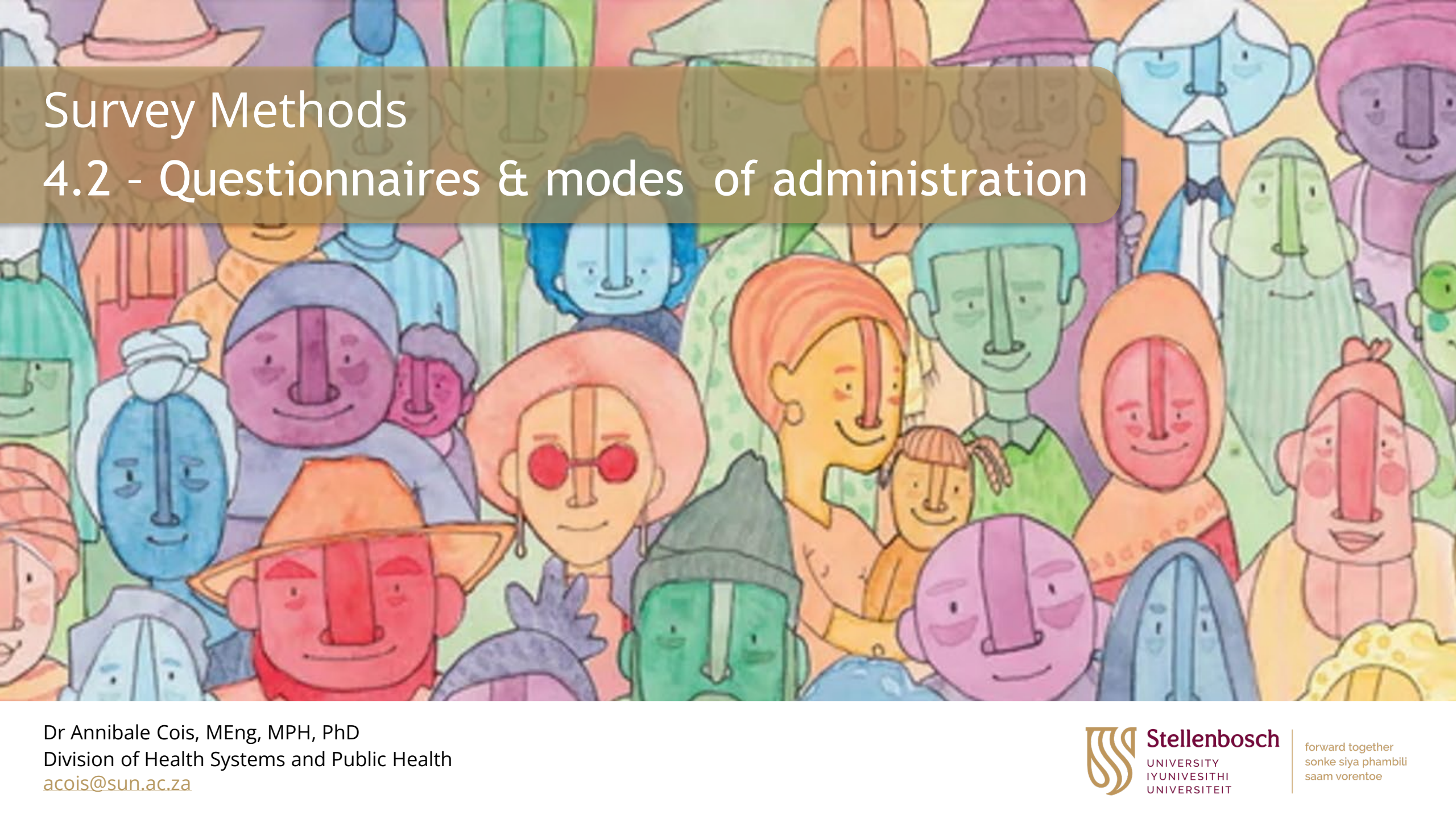
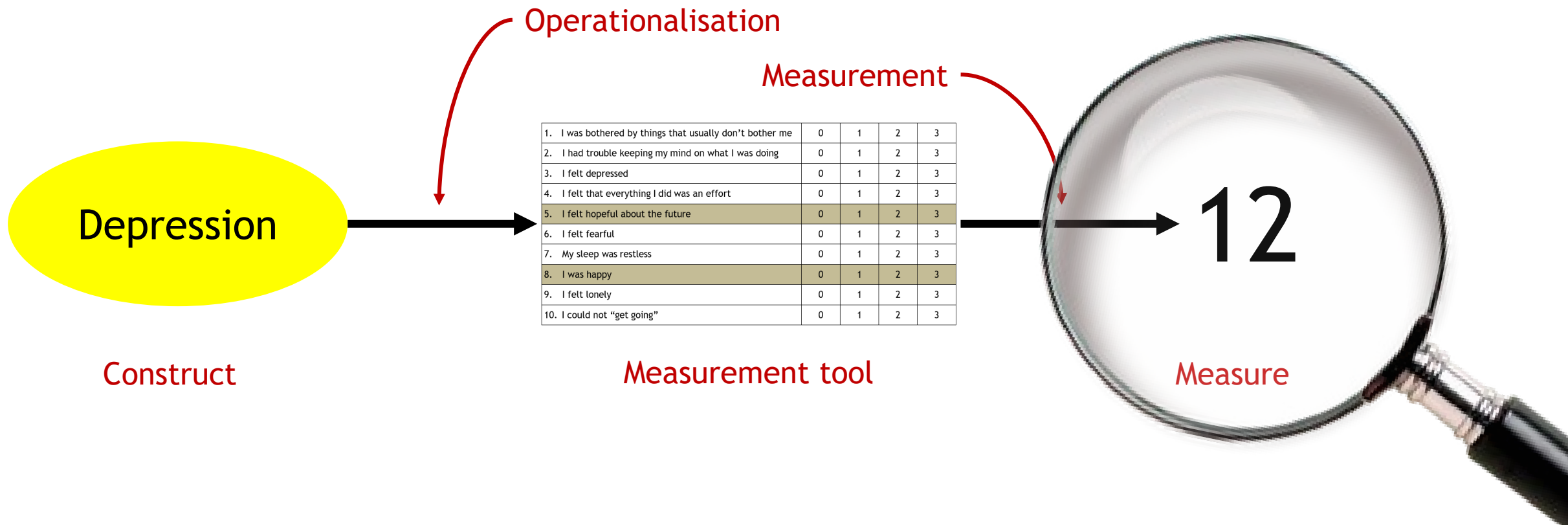




Survey Methods

4.2 - Questionnaires & modes of administration

- Question content, format and wording
- Questionnaire layout and response format
- Self-administered/Interviewer administered Questionnaires
- Online surveys
- Digital data collection systems



How do we get those ‘numbers’?

1. I was bothered by things that usually don't bother me	0	1	2	3
2. I had trouble keeping my mind on what I was doing	0	1	2	3
3. I felt depressed	0	1	2	3
4. I felt that everything I did was an effort	0	1	2	3
5. I felt hopeful about the future	0	1	2	3
6. I felt fearful	0	1	2	3
7. My sleep was restless	0	1	2	3
8. I was happy	0	1	2	3
9. I felt lonely	0	1	2	3
10. I could not "get going"	0	1	2	3

Questionnaires
(self-report)



Direct
measurements

Type of questions

Closed-ended questions: Pre-coded answers, and respondents must choose between the available options. This provides precise, easy to analyse information but restricts options/choices, which may make the data less rich.

Other or not “other”?

Open-ended questions: No pre-coded answers so that respondents can answer however they like. This is an advantage when we lack knowledge of likely answers and must be used when it is not possible to collect data in pre-coded formats. However, open-ended questions will require coding—which may be difficult and time-consuming—before data analysis is possible.

Field coding: Between open-ended and closed-ended questions, field coding is when respondents answer in an open format verbatim, and then the interviewer codes the answers into predetermined response categories.

- This technique can be useful if there is a reason to hide the response options and may be used to quantify findings of qualitative surveys such as focus group discussions (FGDs). It does require, however, that the enumerator is skilled in assigning narrative responses to codes.



Adapted from:

<https://www.povertyactionlab.org/resource/survey-design>

Formulating “good” questions: the principles

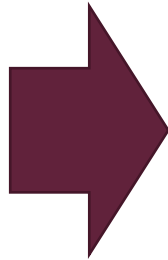
The development of survey questions should be guided by practical experience with the same or comparable questions in other studies, scientific experiments that test alternative ways of asking questions and the magnitude of the errors that result, and theoretical expectations about what ways of asking a question will produce what outcomes based on cognitive psychology and other conceptual approaches to total survey design.

(Aday and Cornelius 2006:194)

Formulating “good” questions: criteria

The criteria for evaluating survey questionnaires include the clarity, balance, and length of the survey questions themselves; the comprehensiveness or constraints of the responses implied or imposed by these questions; the utility of the instructions provided for answering them; the order and context in which they are integrated into the survey questionnaire; and the response errors or effects that result from how the questionnaire is ultimately designed, as well as from the characteristics and behavior of the respondents who answer it, the interviewers who administer it to them, and the mode of data collection employed.

(Aday and Cornelius 2006:194)



1. I was bothered by things that usually don't bother me	0	1	2	3
2. I had trouble keeping my mind on what I was doing	0	1	2	3
3. I felt depressed	0	1	2	3
4. I felt that everything I did was an effort	0	1	2	3
5. I felt hopeful about the future	0	1	2	3
6. I felt fearful	0	1	2	3
7. My sleep was restless	0	1	2	3
8. I was happy	0	1	2	3
9. I felt lonely	0	1	2	3
10. I could not "get going"	0	1	2	3



- Clarity
- Balance
- Length
- Comprehensiveness
- Constraints
- Instructions
- Mode of data collection



Constraints

Table 1
Reported Daily TV Consumption as a Function of Response Alternatives

Low-frequency alternatives	Daily consumption	High-frequency alternatives	Daily consumption
Up to ½ hour	7.4%	Up to 2½ hours	62.5%
½ hour to 1 hour	17.7%	2½ hours to 3 hours	23.4%
1 hour to 1½ hours	26.5%	3 hours to 3½ hours	7.8%
1½ hours to 2 hours	14.7%	3½ hours to 4 hours	4.7%
2 hours to 2½ hours	17.7%	4 hours to 4½ hours	1.6%
More than 2½ hours	16.2%	More than 4½ hours	0.0%

Note. $N = 132$. From "Response Categories: Effects on Behavioral Reports and Comparative Judgments," by N. Schwarz, H. J. Hippler, B. Deutsch, & F. Strack, 1985, *Public Opinion Quarterly*, 49, p. 391. Copyright 1985 by The University of Chicago Press. Adapted with permission.

(Table form: Schwarz 1999)

Constraints



Young America's
dilemma:
Shall I be wise and great,
or rich and powerful?

Dalrymple, Louis, 1866-1905. Poster form 1901. Library of Congress *Catalog*.

<https://www.loc.gov/pictures/item/2010651418/>

Constraints

- Double-barreled questions:
- ask about multiple things in a single question
 - How satisfied are you with the ice cream flavors and toppings at JP Licks?
...What if I am satisfied with flavors but not toppings?



Length

Effect of questionnaire length, personalisation and reminder type on response rate to a complex postal survey: randomised controlled trial

[Shannon Sahlqvist](#) , [Yena Song](#), [Fiona Bull](#), [Emma Adams](#), [John Preston](#), [David Ogilvie](#) & [the iConnect consortium](#)

BMC Medical Research Methodology **11**, Article number: 62 (2011) | [Cite this article](#)

19k Accesses | **136** Citations | **1** Altmetric | [Metrics](#)

Results

An overall response rate of 17% was achieved. Participants who received the short version of the questionnaire were more likely to respond (OR = 1.48, 95% CI 1.06 to 2.07).

...Item non-response was relatively low, but was significantly higher in the long questionnaire than the short (9.8% vs 5.8%; $p = .04$).

Conclusions

In contrast to some previous studies of shorter questionnaires, this trial found that shortening a relatively lengthy questionnaire significantly increased the response.

Researchers should consider the trade off between the value of additional questions and a larger sample.

More specific guidelines

Questions About Health

(Aday and Cornelius 2006:221-244)

Questions About Knowledge and Attitudes

(Aday and Cornelius 2006:245-267)

(Groves et al. 2009:248:251)

Questions About Demographics and Behavior

(Aday and Cornelius 2006:268:287)

(Groves et al. 2009:243:248)

Layout: **do** and

don't

READ YOURSELF!

- 1a. Assign numbers to each question.
- 1b. Use letters to indicate subparts of a question when it has more than one part.
 - 1a. If a center to treat people with AIDS was going to be set up in your neighborhood, would you *favor* or *oppose* it?
Favor (Skip to Q. 2).....1
Oppose.....2
(If "oppose"):
 - 1b. Why is that? _____
2. Use a vertical response format for closed-end responses.
White.....1
African American.....2
Hispanic.....3
3. Use numerical codes for closed-end responses.
In general, would you say that your health is . . .
(Read categories and circle number for answer.)
Excellent.....1
Good.....2
Fair.....3
Poor?.....4
4. Use consistent numerical codes and formats.
Yes.....1
No.....2
Don't know.....8

- Don't* leave off the question number.
Don't leave off the letter for subparts of a question.
- If a center to treat people with AIDS was going to be set up in your neighborhood, would you *favor* or *oppose* it?
Favor (Skip to Q. 2)1
Oppose.....2
(If "oppose"):
Why is that? _____
- Don't* list closed-end responses horizontally.
White....1 African American....2 Hispanic....3
- Don't* use alphabetical codes or blank lines to place X or check on, for closed-end responses.
In general, would you say that your health is . . .
- Excellent _____
Good _____
Fair _____
Poor _____
- Don't* use different numerical codes and formats for comparable responses to different questions.
1 Yes 2 No 3 Don't know
1. Yes
2. No
8. Don't know

(Table adapted from: Aday and Cornelius 2006:289+)

do

don't

READ YOURSELF!

5. Align response codes.

Yes.....1
No.....2
Don't know.....8

White.....1
African American.....2
Hispanic.....3

6. Provide clear instructions for open-ended items.

What was your blood pressure the last time you had it checked?

RECORD HIGH VALUE: _____

(systolic reading)

RECORD LOW VALUE: _____

(diastolic reading)

7. Provide clear special instructions.

(Ask males only):

Did you use a condom?

8. Provide clear skip instructions.

8a. Do you smoke cigarettes?

Yes (Ask Q. 8b).....1

No (Skip to Q. 9).....2

8b. How many cigarettes do you smoke per day on average?

RECORD NUMBER OF CIGARETTES: _____

9. Phrase full and complete questions.

What is your age? or

What is your date of birth?

Don't vary alignment of response codes on a page.

Yes.....1

No.....2

Don't know.....8

White.....1

African American.....2

Hispanic.....3

Don't just leave a space with no instructions for the answer.

What was your blood pressure the last time you had it checked?

Don't have instructions about how to answer questions in same typeface and format as question.

Ask males only:

Did you use a condom?

Don't leave out explicit skip instructions.

8a. Do you smoke cigarettes?

Yes.....1

No.....2

8b. How many cigarettes do you smoke per day on average? _____

Don't simply use words or headings to elicit information from respondents.

Age? _____

do

don't

READ YOURSELF!

10. Use a forced-choice format for a list.

Should an employer be allowed to require job applicants to be medically tested for . . . (Circle answer for yes or no to each.)

	Yes	No
a. Sexually transmitted diseases (STDs)?.....	1	2
b. Using illegal drugs?.....	1	2
c. High blood pressure?	1	2
d. Having AIDS virus?.....	1	2

11. Use a column format for a series with the same response categories.

	Strongly agree	Agree	Disagree	Strongly disagree
a	1	2	3	4
b	1	2	3	4
c	1	2	3	4
d	1	2	3	4
e	1	2	3	4
f	1	2	3	4
g	1	2	3	4

Don't ask respondent to indicate "all that apply" if he or she could indicate more than one response.

Should an employer be allowed to require applicants to be medically tested for . . . (Circle all that apply.)

a. Sexually transmitted diseases (STDs)?.....	1
b. Using illegal drugs?.....	2
c. High blood pressure?.....	3
d. Having AIDS virus?.....	4

Don't repeat a string of questions with the same response categories.

a. Male homosexuals are disgusting.	
Strongly agree.....	1
Agree.....	2
Disagree.....	3
Strongly disagree.....	4
b. Male homosexuality is a natural expression of sexuality.	
Strongly agree.....	1
Agree.....	2
Disagree.....	3
Strongly disagree.....	4

do

don't

READ YOURSELF!

12. Use a column format for a series with comparable skip patterns.

Don't fail to clearly link a series of questions to subsequent dependent items.

For each provider
marked yes in Q. 1:

1. Did you go to see any of the following providers in the past twelve months, from (date) to (date)? (Read list and circle number for yes or no.)

	Yes	No
a. Dentist?	1	2
b. Chiropractor?	1	2
c. Psychotherapist?	1	2

2. How many times did you go see a (provider) during this period? (Record number.)

____ times
____ times
____ times

1. Did you go to see any of the following providers in the past twelve months, from (date) to (date)? (Read list and circle number for yes or no.)

	Yes	No
a. Dentist?	1	2
b. Chiropractor?	1	2
c. Psychotherapist?	1	2

2. How many times did you go see this type of provider during this period? (Record number.)

____ times

13. Put all parts of a question on the same page.

Don't split a question between pages, particularly when skip instructions are part of the question.

14. Allow plenty of space on the questionnaire.

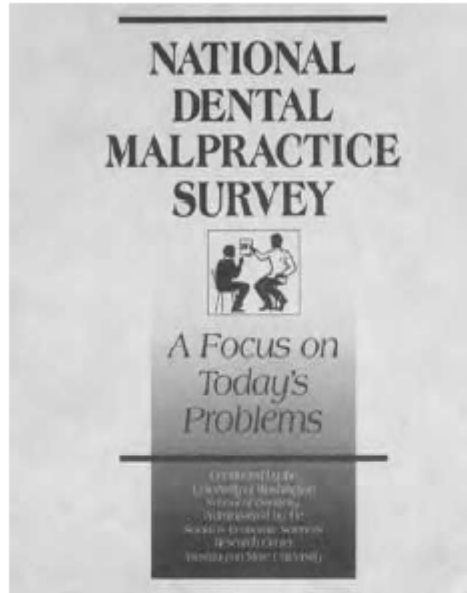
Don't crowd the questions and space for recording the answers.

15. Carefully consider the appearance of the questionnaire.

Don't just start the questions on page 1 without introducing the study, identifying the sponsoring organization, and so on.

do

15. Carefully consider the appearance of the questionnaire.



16. End the questionnaire with a thank you.

Your contribution to this effort is greatly appreciated.
Please enclose the questionnaire in the stamped
self-addressed envelope and mail to:

.....
.....
.....

17. Consider how the data will be processed.

don't

Don't just start the questions on page 1 without introducing the study, identifying the sponsoring organization, and so on.

Don't forget to thank the respondent.

THE END.

Don't fail to anticipate how you will code and process the data as you design the questionnaire.

READ YOURSELF!

Mode of administration



Survey Methods 2022

Face-to-face

Telephonic

Email / Post

Online

Interviewer-administered

Self-administered

CATI

CAPI

Paper-and-pencil

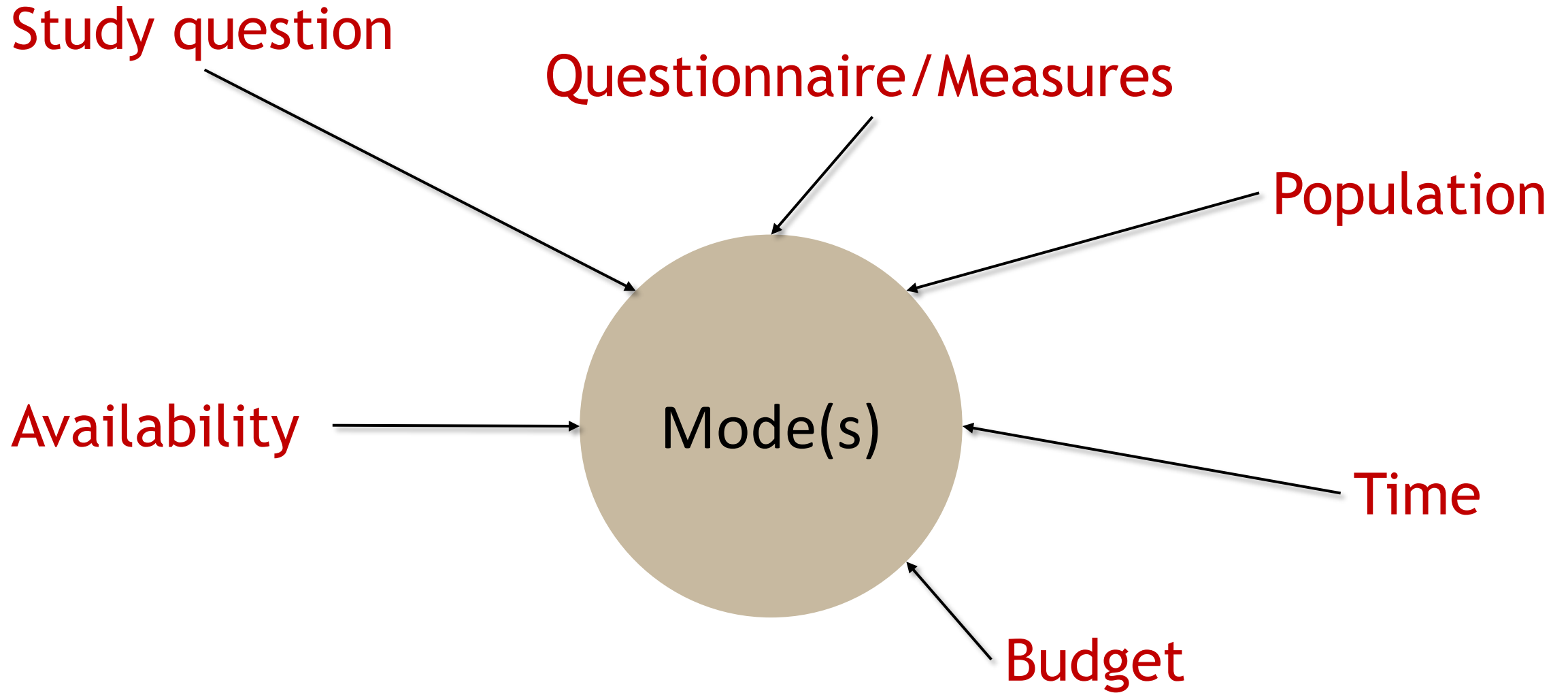
CASI



VIDEOCALLS



Which mode?



In-person

- Pros:
 - Higher response rates; decreases the number of “Don’t know” and “No answer”
 - Can access hard-to-reach populations (e.g. senior citizens)
 - Can be done in a variety of settings
 - Can involve all 5 senses (e.g. taste test)
- Cons:
 - More expensive and time consuming
 - Interviewer error
 - Can be complex



Telephone

- Pros:

- Higher response rates; decreases the number of “Don’t know” and “No answer”
- Lower cost and less time than in-person interviews
- Can be computer-assisted

- Cons:

- Unlisted numbers
- Cell phones
- Telemarketing ruined it for everyone

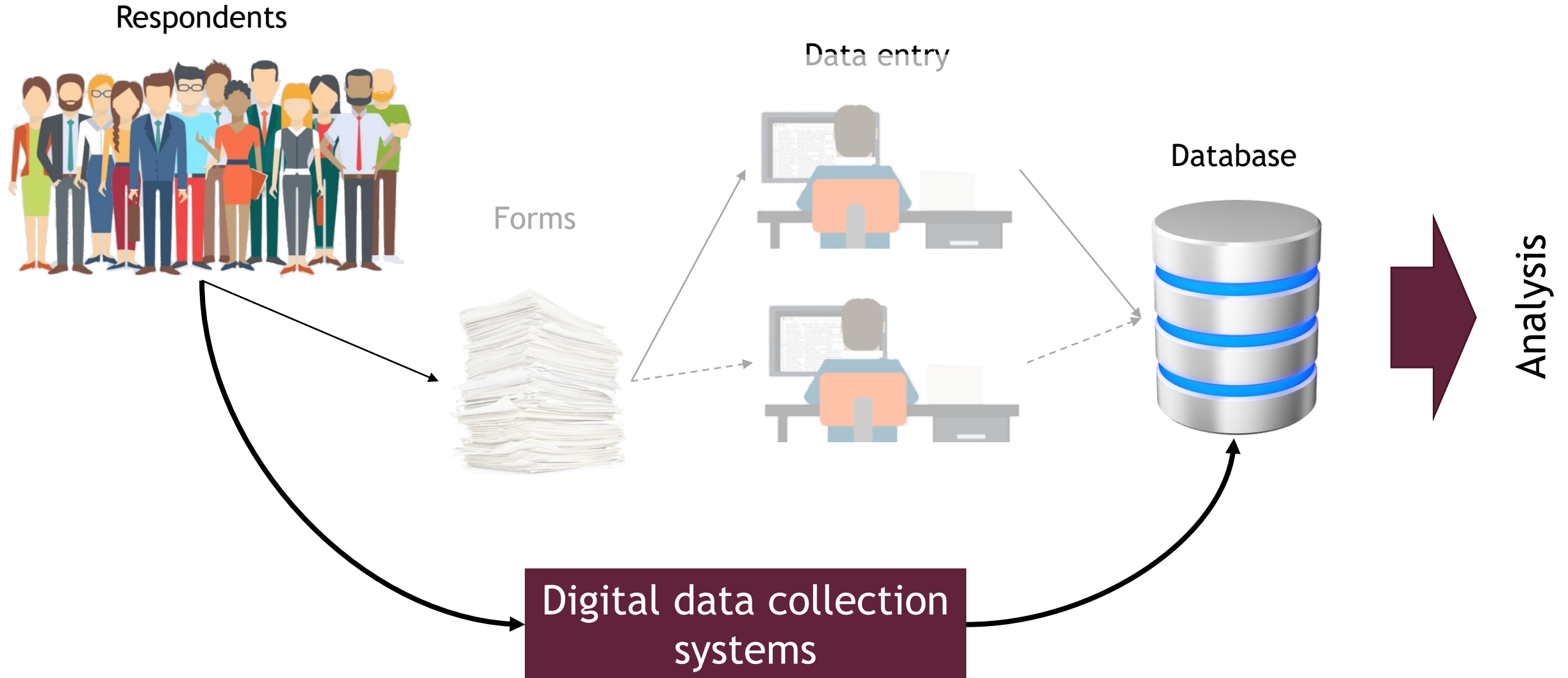
Mail/email

- Pros:
 - Large samples
 - Cheaper than interviews
 - Respondent does survey on own time
- Cons
 - More expensive than online
 - Low response rates
 - Incomplete answers (illegible for mail surveys)

Online

- Pros:
 - Inexpensive, least time-consuming
 - Automatic data entry
 - Can easily merge with additional data
- Cons:
 - Lower response rates than in-person
 - Respondents must have access to computer, use an email address
 - Technical errors

Digital data collection systems



Digital data collection systems

A good investment, even
for small surveys!

Planning



Execution



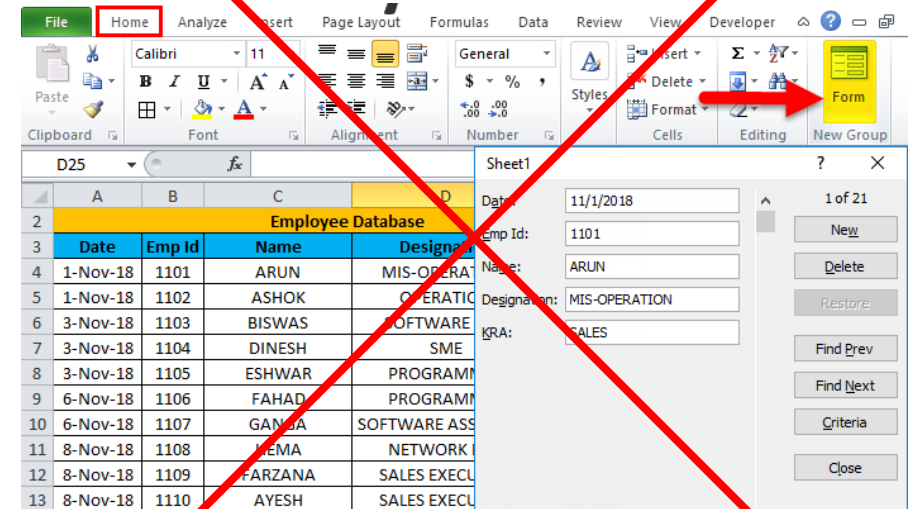
Which data collection systems

Specialised system



<https://redcap.sun.ac.za/index.php?action=myprojects>

Spreadsheets



REDCap Features

Build online surveys and databases quickly and securely in your browser

- Create and design your project using a secure login from any device. No extra software required. Access from anywhere, at any time.

Fast and flexible - Go from project creation to starting data collection in less than one day. Customizations and changes are possible any time, even after data collection has begun.

Advanced instrument design features - Auto-validation, calculated fields, file uploading, branching/skip logic, and survey stop actions.

Diverse and flexible survey distribution options - Use a list of email addresses or phone numbers for your survey respondents and automatically contact them with personalized messages, and track who has responded. Or create a simple link for an anonymous survey for mass email mailings, to post on a website, or print on a flyer.

Data quality - Use field validation, branching/skip logic, and Missing Data Codes to improve and protect data quality during data entry. Open data queries to automatically identify and resolve discrepancies and other issues real-time.

Custom reporting - Create custom searches for generating reports to view aggregate data. Identify trends with built-in basic statistics and charts.

Export data to common analysis packages - Export your data as a PDF or as CSV data for easy analysis in SAS, Stata, R, SPSS, or Microsoft Excel.

Secure file storage and sharing - Upload and share any type of file with anyone in the world through the File Repository feature or Send-It tool. Also works with exports and other built-in file uploading features.

Data-based triggers and alerts - Send real-time alerts and notifications to your team or other stakeholders via email, text, or phone based on certain data being entered or specific questions having a particular answer.

Connect to other resources - Use built-in features (API) to move data to/from your project. Build your own custom software development features to connect your project to other systems.



Log In

Log in using your Stellenbosch University username and password

Dear SU Researcher, Please find REDCap onboarding guidance [here](#), and an introductory presentation [here](#). If you have remaining questions after you've consulted the guidance documents, feel free to contact redcap@sun.ac.za

The SU instance of REDCap was successfully upgraded to version 12.0.11 on 22 January 2022. Please find a listing of some new features [here](#).

Please log in with your user name and password. If you are having trouble logging in, please contact [REDCap Administrator](#).

Username:

Password:

Log In

A case against using spreadsheets as data collection/management/tools

Limited scalability

Modest set of data operations

No Error Control

Little Reusability

Proprietary format,
proprietary code

Lack of Automation

Difficult Collaboration

Low level of data protection

Summary questions

- Evaluation criteria for question content, format and wording
- Evaluation criteria for questionnaire layout and response format
- Modes of administration and main pros/cons
- Digital data collection systems

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Thank you

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